

# SYSTEM DESIGN TRICKS

## For Wireless Communication



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**T**he launch of Bluetooth 4.0 was a revolution. All of a sudden we had wireless transfer technology that consumed drastically less power while performing better in terms of range and capacity. As for Wi-Fi, we have the ever-trustworthy 802.11x standards with constant increase in speed over the years—802.11ah is an improvement over the earlier 802.11ad, which itself is under development. It can also penetrate walls and other barriers considerably better than previous Wi-Fi standards.

For a holistic picture, we need to look at the improvements over the years. Increase in performance has been coming at the cost of exhaustive power consumption. So let us look at how power can be reduced in devices while increasing performance. Bluetooth 4.0 or Bluetooth Low Energy (BLE) allow reduction in power without any compromise in performance.

### Improving performance with lesser power

An area of significant improvement has been the usage of modules with power-saving features, especially when the system is idle. In such modules, standby power consumption typically ranges between 1.9mA and 5.7mA, which is ideal for battery-powered systems.

Modules that go into sleep mode when wireless transmission is inactive provide the ability to power individual blocks on the board, like the transmitter or the DSP and, hence, help hugely in improving performance.

Power consumption protocols in BLE consume only a fraction of the power purged by

### Modules over the years

When launched in 1990s, an ARM9 microcontroller handled the entire mobile phone. The processor was used at its full capacity, and the module in an embedded mobile system performed at less than 20 per cent capacity. Often the modules were shipped with only certain parts of the chipset connected to the external package, rendering its core unusable.

While there is plenty of processing capacity left in wireless modules, most modules we have today are provided as black-boxes, with no insight into the internal operation to access the extra capacity. Which leaves a big question unanswered: Which modules are actually available as open to access the extra power?

earlier versions. This is not done by sacrificing on signal strength, but by altering the duty cycle. It is optimised for small bursts of data where continuous use actually makes the device consume more power. Transmitting via bursts does not cause loss in data but allows you to save power. This data is then transmitted via the antenna, which is another major player in wireless performance. A conventional finite ground coplanar waveguide, for example, can be modified to obtain wide-band response at desired frequencies.

### The receiver plays a big role.

Wireless performance can be directly increased by increasing transmitting power, but that would not be productive. Increasing the transmitted power will get you longer range but could also lead to multichannel signal interference. Also, load on a mobile battery would cause it to be consumed in a matter of few hours.

Sensitivity of the wireless receiver can, however, be increased to make a change in this regard. Theoretically speaking, increasing the sensitivity by 6dB doubles the range for reception.

Fig. 1: Wireless charging

